



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

CS220: Database Systems

Class: BSAI – 2A

Lab 11: Open Ended Lab

CLO-2: Formulate SQL queries to retrieve information from a relational database.

CLO 3: Build SQL (DDL &DML) queries to retrieve information from a relational database.

Date: 16 April 2026

Time: 10:00 - 13:00



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Introduction

This lab is Open Ended Lab a complete scenario is given and you are required to propose the solutions as per your understandings.

Objectives

Objective of this lab is to understand the scenario, make ERD, write DDL , populate data through DML and Execute DRL for reports.

Animals Well-being Management System

Akmal, a loyal client of Dist Haywanaat Shifa Khana “DHSK”, arrives with his one of the **golden retrievers (Dogs)**, **Tommy**, for Tommy's scheduled biannual vaccination. This time, Tommy requires an oral vaccination that involves Treatment of **local anesthesia**. The veterinarian accesses Tommy's comprehensive medical history and notes the need for the oral vaccination. To ensure Tommy's comfort, the vet administers a local anesthesia treatment before proceeding with the **oral vaccination** process.

During the visit, various treatments and medications are provided, including the **oral vaccine**, **the local anesthesia**, and any additional medications prescribed for Tommy's well-being. Each treatment, medication, its **dosage**, and **price** are very well documented.

Post-treatment, the front desk staff compile an itemized bill detailing the treatment and medications. The **total cost** of the visit, including the oral vaccination, local anesthesia, and any additional services rendered, arriving at a total of 2454/- rupees for the comprehensive treatment.

Consider the following Business Rules:

1. A customer can have many pets but must have at least one.
2. A pet must be assigned to one and only one customer.
3. A pet can have one or more treatments per visit but must have at least one.
4. A treatment can have one or more medications.
5. A bill must have details of the treatments including costs.



Task:

1: Analyze the scenario to understand the working of a DHSK .

DHSK system manages:

- Customers (owners)
- Pets (each belongs to one owner)
- Visits (each visit = appointment)
- Treatments (done in a visit)
- Medications (used in treatments)
- Billing (cost calculation)

We have 6 main tables in this system namely Customers, Pets, Visits, Treatments, Medications and billings each having its own attributes and their relationships with other tables. Next we calculate the billing which is the total cost of treatments, medications and visit fees for each of the pets.

2: Identify Entities, attributes and constraints (identify appropriate related attributes)

- ✚ I have highlighted in yellow most of the attributes I could find in the system example description.
- ✚ For each table I have mentioned their respective primary keys and foreign keys if any.
- ✚ I made 2 junction tables Visit_Treatment & Treatment_Medicine because there are 2 cases in this systems where both tables are having a M:N relationship and hence we require relationship tables.
- ✚ The junction tables attributes are created from certain properties e.g. they must contain primary keys of both the tables as attributes which will act like foreign keys and that they will have their own independent primary keys as well.



ENTITY-ATTRIBUTES TABLE

Entity Name	Attributes	Primary Key	Foreign Key(s)
Customer	customer_id, name, phone	customer_id	—
Pet	pet_id, name, type, breed, customer_id	pet_id	customer_id → Customer(customer_id)
Visit	visit_id, pet_id, visit_date	visit_id	pet_id → Pet(pet_id)
Treatment	treatment_id, name, cost	treatment_id	—
Medication	medicine_id, name, price	medicine_id	—
Visit_Treatment	vt_id, visit_id, treatment_id	vt_id	visit_id → Visit(visit_id), treatment_id → Treatment(treatment_id)
Treatment_Medicine	tm_id, treatment_id, medicine_id, dosage	tm_id	treatment_id → Treatment(treatment_id), medicine_id → Medication(medicine_id)
Billing	bill_id, visit_id, total_cost, bill_date	bill_id	visit_id → Visit(visit_id)

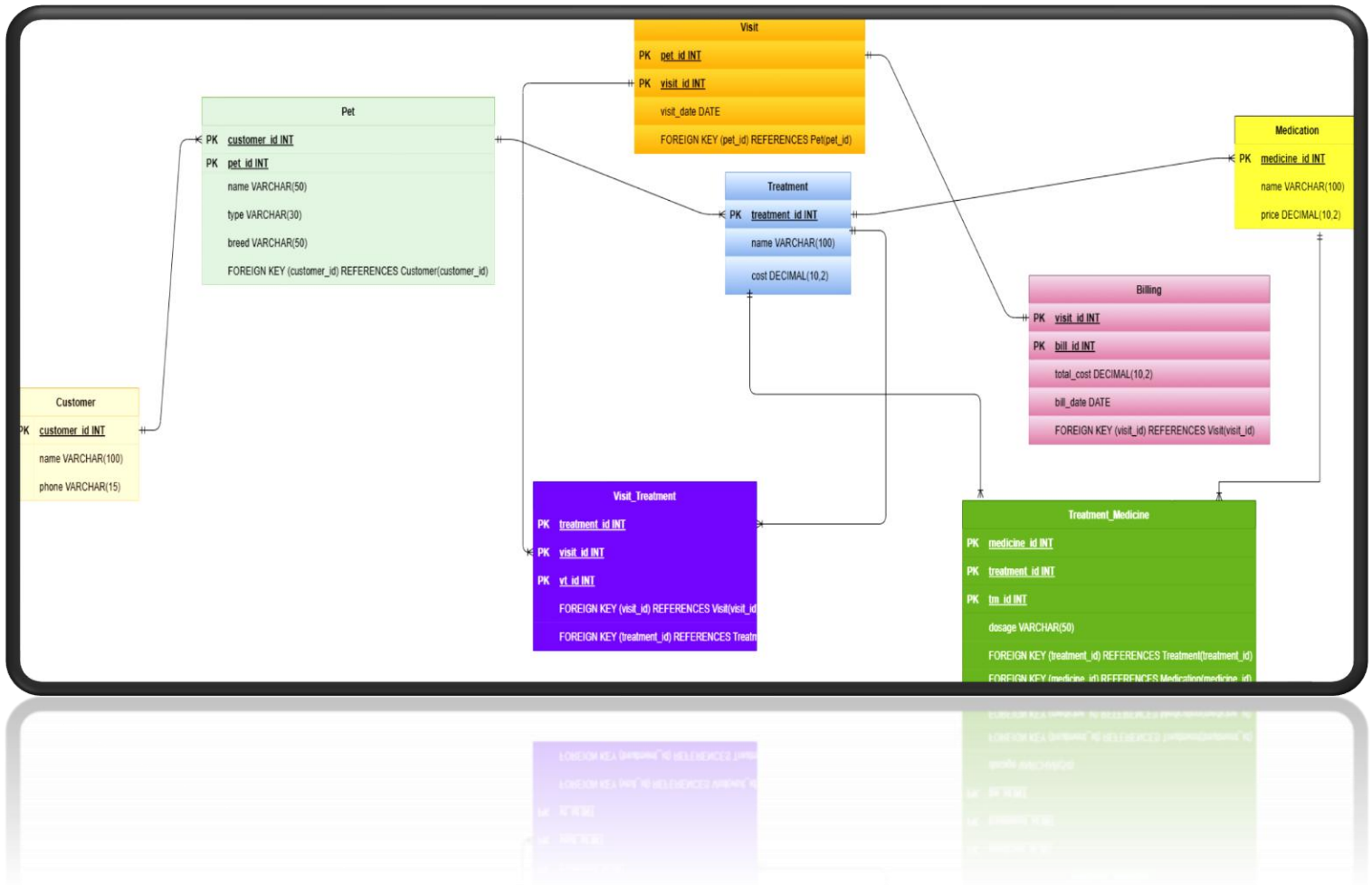
3: Design ERD to meet the requirements



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ERD DIAGRAM:





4: Implement it into an executable relational database

SQL QUERY:

```
use lab11;
```

```
CREATE TABLE Customer (  
    customer_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    phone VARCHAR(15)  
);
```

```
CREATE TABLE Pet (  
    pet_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    type VARCHAR(20),  
    breed VARCHAR(50),  
    customer_id INT,  
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)  
);
```

```
CREATE TABLE Visit (  
    visit_id INT PRIMARY KEY,  
    pet_id INT,  
    visit_date DATE,  
    FOREIGN KEY (pet_id) REFERENCES Pet(pet_id)  
);
```

```
CREATE TABLE Treatment (  
    treatment_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    cost INT  
);
```

```
CREATE TABLE Medication (  
    medicine_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    price INT  
);
```



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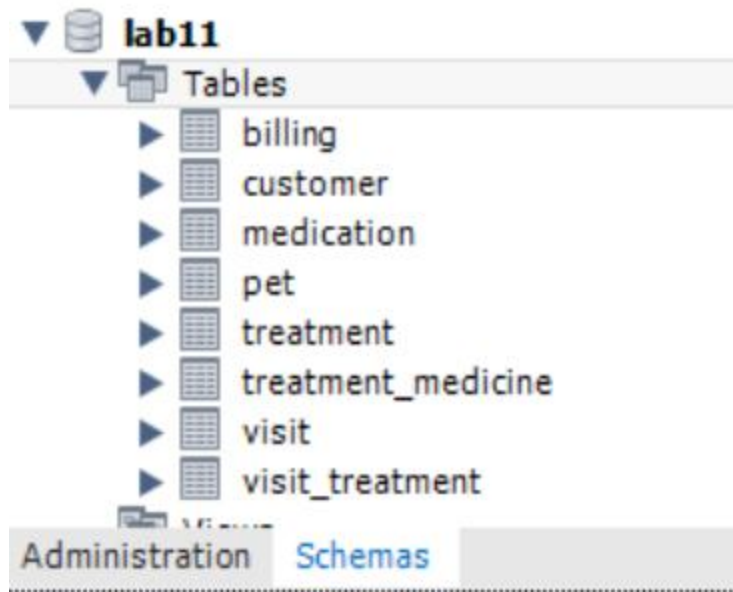
```
CREATE TABLE Visit_Treatment (  
    vt_id INT PRIMARY KEY,  
    visit_id INT,  
    treatment_id INT,  
    FOREIGN KEY (visit_id) REFERENCES Visit(visit_id),  
    FOREIGN KEY (treatment_id) REFERENCES Treatment(treatment_id)  
);
```

```
CREATE TABLE Treatment_Medicine (  
    tm_id INT PRIMARY KEY,  
    treatment_id INT,  
    medicine_id INT,  
    dosage VARCHAR(20),  
    FOREIGN KEY (treatment_id) REFERENCES Treatment(treatment_id),  
    FOREIGN KEY (medicine_id) REFERENCES Medication(medicine_id)  
);
```

```
CREATE TABLE Billing (  
    bill_id INT PRIMARY KEY,  
    visit_id INT UNIQUE,  
    total_amount INT,  
    bill_date DATE,  
    FOREIGN KEY (visit_id) REFERENCES Visit(visit_id)  
);
```

OUTPUT:

✓	4	15:37:31	use lab11	0 row(s) affected	0.000 sec
✓	5	15:37:31	CREATE TABLE Customer (customer_id INT PRIMARY KEY, name V...	0 row(s) affected	0.266 sec
✓	6	15:37:31	CREATE TABLE Pet (pet_id INT PRIMARY KEY, name VARCHAR(5...	0 row(s) affected	0.062 sec
✓	7	15:37:31	CREATE TABLE Visit (visit_id INT PRIMARY KEY, pet_id INT, visit...	0 row(s) affected	0.047 sec
✓	8	15:37:31	CREATE TABLE Treatment (treatment_id INT PRIMARY KEY, name ...	0 row(s) affected	0.032 sec
✓	9	15:37:31	CREATE TABLE Medication (medicine_id INT PRIMARY KEY, name ...	0 row(s) affected	0.015 sec
✓	10	15:37:31	CREATE TABLE Visit_Treatment (vt_id INT PRIMARY KEY, visit_id I...	0 row(s) affected	0.047 sec
✓	11	15:37:31	CREATE TABLE Treatment_Medicine (tm_id INT PRIMARY KEY, trea...	0 row(s) affected	0.047 sec
✓	12	15:37:32	CREATE TABLE Billing (bill_id INT PRIMARY KEY, visit_id INT UNIQ...	0 row(s) affected	0.031 sec



5: Populate the database with 3 owners, 5 pets, 3 treatments, 7 medicines, 1 old visit and 1 current visit

SQL QUERY:

-- Customers

```
INSERT INTO Customer VALUES (1,'Akmal','0300');
```

```
INSERT INTO Customer VALUES (2,'Mike','0311');
```

```
INSERT INTO Customer VALUES (3,'Jack','0322');
```

-- Pets

```
INSERT INTO Pet VALUES (1,'Tommy','Dog','Golden Retriever',1);
```

```
INSERT INTO Pet VALUES (2,'Heera','Chicken','Punjab',1);
```

```
INSERT INTO Pet VALUES (3,'Michi','Chicken','Egyptian',2);
```

```
INSERT INTO Pet VALUES (4,'Narp','Chick','Hen',3);
```

```
INSERT INTO Pet VALUES (5,'Buhugs bunny','Rabbit','Pakistani',3);
```

-- Visits

```
INSERT INTO Visit VALUES (1,1,'2025-01-01'); -- old
```

```
INSERT INTO Visit VALUES (2,1,'2026-04-16'); -- current
```

-- Treatments



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```
INSERT INTO Treatment VALUES (1,'Oral Vaccination',1000);  
INSERT INTO Treatment VALUES (2,'Local Anesthesia',800);  
INSERT INTO Treatment VALUES (3,'Checkup',500);
```

-- Medicines

```
INSERT INTO Medication VALUES (1,'Vaccine A',500);  
INSERT INTO Medication VALUES (2,'Anesthesia Drug',400);  
INSERT INTO Medication VALUES (3,'Painkiller',200);  
INSERT INTO Medication VALUES (4,'Antibiotic',300);  
INSERT INTO Medication VALUES (5,'Vitamin',150);  
INSERT INTO Medication VALUES (6,'Dewormer',250);  
INSERT INTO Medication VALUES (7,'Supplement',100);
```

-- Visit_Treatment

```
INSERT INTO Visit_Treatment VALUES (1,1,3);  
INSERT INTO Visit_Treatment VALUES (2,2,1);  
INSERT INTO Visit_Treatment VALUES (3,2,2);
```

-- Treatment_Medicine

```
INSERT INTO Treatment_Medicine VALUES (1,1,1,'5ml');  
INSERT INTO Treatment_Medicine VALUES (2,2,2,'2ml');  
INSERT INTO Treatment_Medicine VALUES (3,3,3,'1ml');  
INSERT INTO Treatment_Medicine VALUES (4,3,4,'9ml');  
INSERT INTO Treatment_Medicine VALUES (5,1,5,'12ml');  
INSERT INTO Treatment_Medicine VALUES (6,2,6,'3ml');  
INSERT INTO Treatment_Medicine VALUES (7,2,7,'19ml');
```

-- Billing

```
INSERT INTO Billing VALUES (1,1,500,'2025-01-01');  
INSERT INTO Billing VALUES (2,2,2454,'2026-04-16');
```



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OUTPUT:

Action Output				
#	Time	Action	Message	
✓ 14	15:49:06	INSERT INTO Customer VALUES (2,'Mike','0311')	1 row(s) affected	
✓ 15	15:49:06	INSERT INTO Customer VALUES (3,'Jack','0322')	1 row(s) affected	
✓ 16	15:49:06	INSERT INTO Pet VALUES (1,'Tommy','Dog','Golden Retriever',1)	1 row(s) affected	
✓ 17	15:49:06	INSERT INTO Pet VALUES (2,'Heera','Chicken','Punjab',1)	1 row(s) affected	
✓ 18	15:49:06	INSERT INTO Pet VALUES (3,'Michi','Chicken','Egyptian',2)	1 row(s) affected	
✓ 19	15:49:06	INSERT INTO Pet VALUES (4,'Narp','Chick','Hen',3)	1 row(s) affected	
✓ 20	15:49:06	INSERT INTO Pet VALUES (5,'Buhugs bunny','Rabbit','Pakistani',3)	1 row(s) affected	
✓ 21	15:49:06	INSERT INTO Visit VALUES (1,1,'2025-01-01')	1 row(s) affected	
✓ 22	15:49:06	INSERT INTO Visit VALUES (2,1,'2026-04-16')	1 row(s) affected	
✓ 23	15:49:06	INSERT INTO Treatment VALUES (1,'Oral Vaccination',1000)	1 row(s) affected	
✓ 24	15:49:06	INSERT INTO Treatment VALUES (2,'Local Anesthesia',800)	1 row(s) affected	
✓ 25	15:49:06	INSERT INTO Treatment VALUES (3,'Checkup',500)	1 row(s) affected	
✓ 26	15:49:06	INSERT INTO Medication VALUES (1,'Vaccine A',500)	1 row(s) affected	
✓ 27	15:49:06	INSERT INTO Medication VALUES (2,'Anesthesia Drug',400)	1 row(s) affected	
✓ 28	15:49:06	INSERT INTO Medication VALUES (3,'Painkiller',200)	1 row(s) affected	
✓ 29	15:49:06	INSERT INTO Medication VALUES (4,'Antibiotic',300)	1 row(s) affected	

PROOFS:

Result Grid					
Filter Rows:					
	pet_id	name	type	breed	customer_id
▶	1	Tommy	Dog	Golden Retriever	1
	2	Heera	Chicken	Punjab	1
	3	Michi	Chicken	Egyptian	2
	4	Narp	Chick	Hen	3
	5	Buhugs bunny	Rabbit	Pakistani	3
•	NULL	NULL	NULL	NULL	NULL



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Result Grid  Filter Rows:				
	bill_id	visit_id	total_amount	bill_date
▶	1	1	500	2025-01-01
	2	2	2454	2026-04-16
✱	NULL	NULL	NULL	NULL

6: Provide the reports through SQL queries for the followings:
a: bill with details of the old visit

SQL QUERY:

```
SELECT v.visit_id, p.name AS pet, t.name AS treatment, t.cost
FROM Visit v
JOIN Pet p ON v.pet_id = p.pet_id
JOIN Visit_Treatment vt ON v.visit_id = vt.visit_id
JOIN Treatment t ON vt.treatment_id = t.treatment_id
WHERE v.visit_id = 1;
```

OUTPUT:

Result Grid  Filter Rows:				
	visit_id	pet	treatment	cost
▶	1	Tommy	Checkup	500



b: bill with details of the current visit

SQL QUERY:

```
SELECT v.visit_id, p.name AS pet, t.name AS treatment, t.cost  
FROM Visit v  
JOIN Pet p ON v.pet_id = p.pet_id  
JOIN Visit_Treatment vt ON v.visit_id = vt.visit_id  
JOIN Treatment t ON vt.treatment_id = t.treatment_id  
WHERE v.visit_id = 2;
```

OUTPUT:

Result Grid		Filter Rows:	
visit_id	pet	treatment	cost
2	Tommy	Oral Vaccination	1000
2	Tommy	Local Anesthesia	800

c: bill having maximum cost with details of the visit

SQL QUERY:

```
SELECT *  
FROM Billing  
WHERE total_amount = (SELECT MAX(total_amount) FROM Billing);
```



OUTPUT:

	bill_id	visit_id	total_amount	bill_date
▶	2	2	2454	2026-04-16
✱	NULL	NULL	NULL	NULL

d: Identify the Pet having maximum distinct treatments along with treatment name.

SQL QUERY:

```
SELECT p.name, COUNT(DISTINCT vt.treatment_id) AS total_treatments
FROM Pet p
JOIN Visit v ON p.pet_id = v.pet_id
JOIN Visit_Treatment vt ON v.visit_id = vt.visit_id
GROUP BY p.pet_id, p.name
ORDER BY total_treatments DESC
LIMIT 1;
```

OUTPUT:

	name	total_treatments
▶	Tommy	3

Deliverables:

Provide a complete report having ERD, DDL, DML, DRL and snapshot of results.